Active sensing explains cortical variability and improves visual information

UCLA Stein Eye Institute · UC Berkeley

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Fixational eye movements: noise, or sensing?



usually: motor noise
↓ here: how the eye senses



usually: neural noise
↓ here: what the eye delivers

During fixation the eye keeps moving the image across the center of gaze, where each neuron sees a patch only about a quarter of a thumbnail wide at arm's length.

Is what looks like noise — in the eye and in cortex — part of how we see?

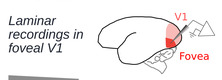
A fingertip, not a camera?

1–5 Most of the "noise" comes from eye motion

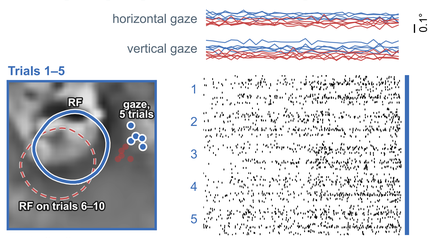
6 It's the image moving on the retina, not a brain signal

7–8 In a model, the motion helps neurons encode fine detail

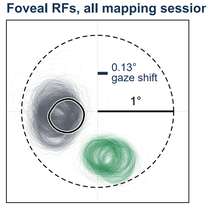
1 Tiny gaze shifts reshape population responses



Dual Purkinje high-resolution eye tracker [2–4]



Foveal RFs, all mapping sessions

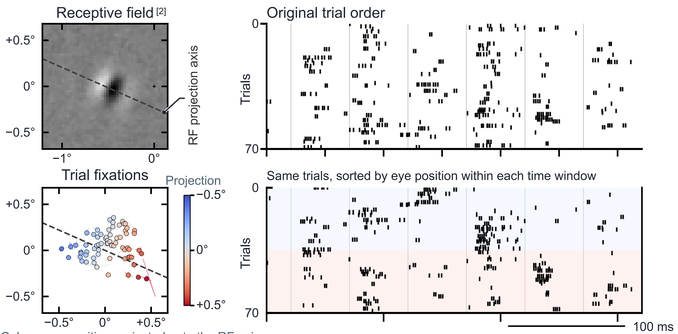


Small changes in fixation drive large changes in population response

Monkey A · Monkey L · this session

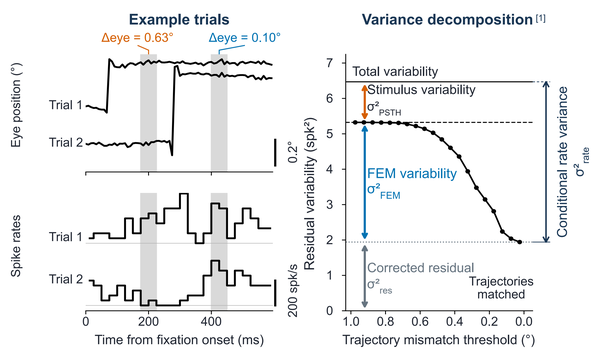
2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

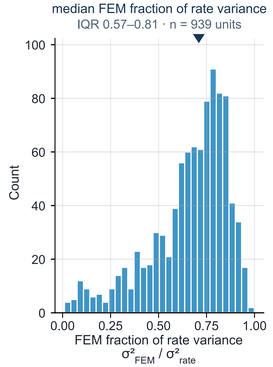
2 One neuron's "noise" tracks the eye



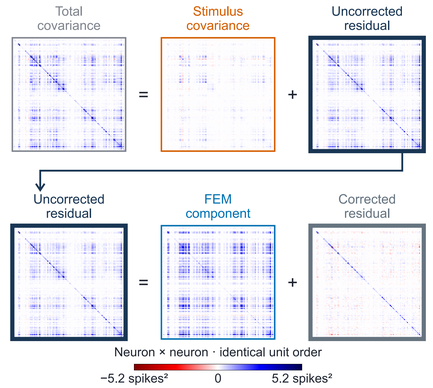
Colors: eye position projected onto the RF axis.

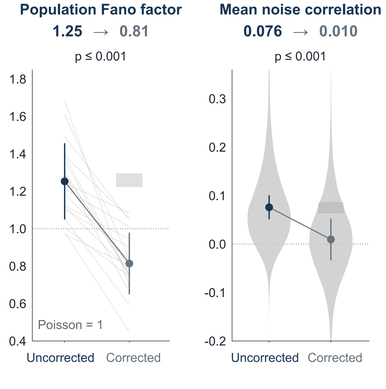
3 Eye movements drive most single-cell rate variance (71%)





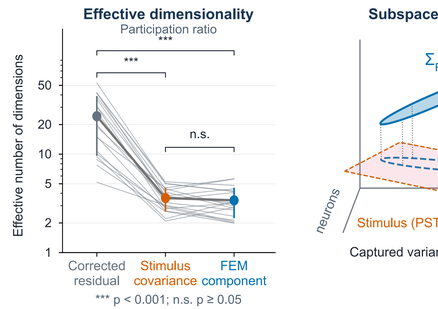
4 Correcting for gaze removes shared noise

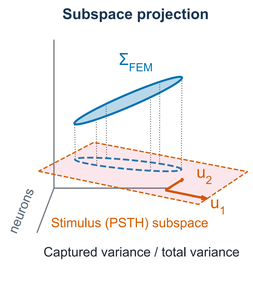


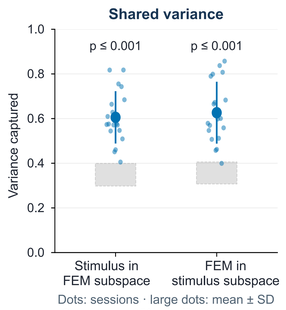


19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions

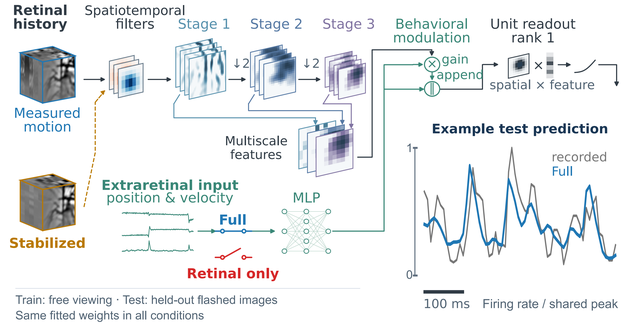


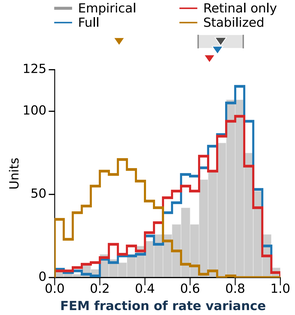




25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

6 Gaze-dependent retinal input predicts single trials



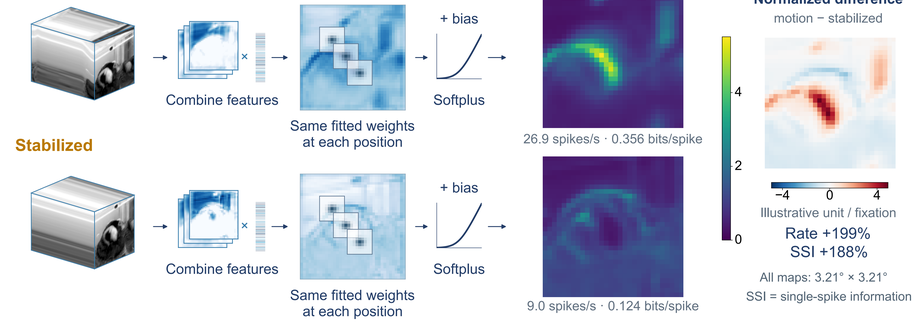


Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions

Triangles: medians; band: empirical ±0.1.

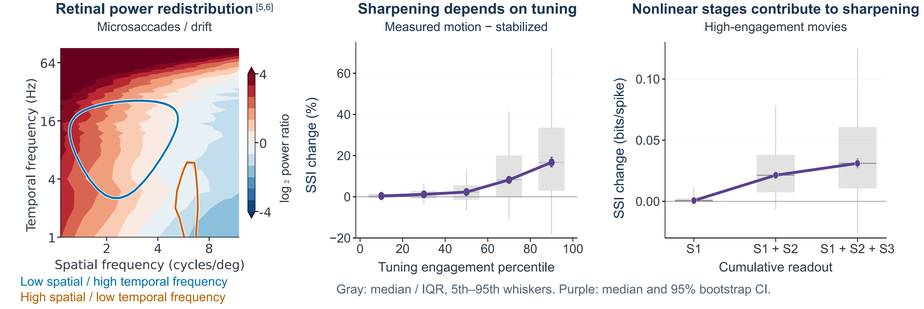
Median fraction of conditional rate variance explained 962 units · 19 sessions	Trial average	Full	Retinal only	Stabilized
	0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).
Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

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